

FORENSIC ANALYSIS REPORT

Image: jo-favorites-usb-2009-12-11.E01 • 2026-02-24 01:44:04 • Model: meta-llama/llama-3.1-8b-instruct

Risk Level: HIGH

EXECUTIVE SUMMARY

The analysis reveals potential timestomping and shadow copy deletion activities, indicating possible anti-forensic techniques used to conceal evidence. Further investigation is required to confirm the extent of these activities and identify potential tools used.

Analysis completed in 10.2 seconds. AI analysis identified 3 findings. Anti-forensic analysis detected 260 potential indicators.

AI ANALYSIS FINDINGS

Finding #1: timestomping	MEDIUM	80% conf
Evidence:	Future timestamps found: 1, Sample future dates: ['USN journal missing']	
Explanation:	The presence of future timestamps, specifically a missing USN journal, suggests that the system's timestamps were manipulated.	
Recommendation:	Verify the integrity of the USN journal and investigate potential timestomping tools used.	
Finding #2: shadow_deletion	HIGH	90% conf
Evidence:	\$Extend metadata exists: False, System Volume Information folder: False	
Explanation:	The absence of the \$Extend metadata and the System Volume Information folder strongly suggests that Volume Shadow Copy Service (VSS) was disabled or manipulated.	
Recommendation:	Investigate potential tools used for shadow copy deletion and verify the integrity of the system's backup mechanism.	
Finding #3: deletion	LOW	40% conf
Evidence:	Orphaned files: 1	
Explanation:	The presence of an orphaned file may indicate that a file was deleted, but this could also be a normal occurrence.	
Recommendation:	Verify the file's history and investigate potential deletion tools used.	

ANTI-FORENSIC ANALYSIS

TIMESTOMPING DETECTION [256 indicators]

Detects manipulation of file timestamps

- Partition 0: Zero timestamp detected: r/r 3: JO FAVORITE (Volume Label Entry) 2009-11-20 09:30:34 (IST) 0000-00-00 00:
 - Partition 0: Zero timestamp detected: r/r 5: ._Trashes 2009-11-17 16:45:26 (IST) 2009-11-20 00:00:00 (IST) 0000-00-00
 - Partition 0: Zero timestamp detected: r/r 6: DSC00009.JPG 2009-11-11 09:16:56 (IST) 2009-11-24 00:00:00 (IST) 0000-00-
 - Partition 0: Zero timestamp detected: d/d 8: .Trashes 2009-11-17 16:45:26 (IST) 2009-11-17 00:00:00 (IST) 0000-00-00 0
 - Partition 0: Zero timestamp detected: + d/d * 133: ._02 2009-11-20 09:25:26 (IST) 2009-11-20 00:00:00 (IST) 0000-00-00
 - Partition 0: Zero timestamp detected: + r/r * 135: ._502 2009-11-20 09:25:28 (IST) 2009-11-20 00:00:00 (IST) 0000-00-0
 - Partition 0: Zero timestamp detected: + r/r * 136: ._502~1 2009-11-17 16:45:28 (IST) 2009-11-17 00:00:00 (IST) 0000-00-
 - Partition 0: Zero timestamp detected: + r/r * 138: ._501 2009-11-17 16:45:28 (IST) 2009-11-17 00:00:00 (IST) 0000-00-0
 - Partition 0: Zero timestamp detected: d/d * 10: .fsevents 2009-11-17 16:45:26 (IST) 2009-11-17 00:00:00 (IST) 0000-00
 - Partition 0: Zero timestamp detected: d/d 13: .Spotlight-V100 2009-11-17 16:45:26 (IST) 2009-11-17 00:00:00 (IST) 0000
 - Partition 0: Zero timestamp detected: + d/d 646: Store-V1 2009-11-17 16:45:26 (IST) 2009-11-17 00:00:00 (IST) 0000-00-
 - Partition 0: Zero timestamp detected: ++ r/r 775: VolumeConfig.plist 2009-11-17 16:45:28 (IST) 2009-11-20 00:00:00 (IST)
 - Partition 0: Zero timestamp detected: ++ d/d 777: Stores 2009-11-17 16:45:28 (IST) 2009-11-17 00:00:00 (IST) 0000-00-0
 - Partition 0: Zero timestamp detected: +++ d/d 1544: BC650BA8-C5B5-4A64-8BDB-18A9533BF1D7 2009-11-17 16:45:28 (IST) 200
 - Partition 0: Zero timestamp detected: ++++ r/r 1669: psid.db 2009-11-20 09:30:34 (IST) 2009-11-20 00:00:00 (IST) 0000-
- ... and 241 more

SHADOW COPY DELETION [Clear]

Evidence of VSS deletion attempts

No indicators found in this category.

HIDDEN DATA STREAMS (ADS) [Clear]

Alternate Data Streams detection

No indicators found in this category.

LOG CLEARING EVIDENCE [2 indicators]

Evidence of log file clearing

- Partition 0: No Windows event log files (.evtx) found in listing
- No Windows Event Logs (.evtx) found in extracted logs directory

REGISTRY TAMPERING [Clear]

Registry modification indicators

No indicators found in this category.

FILE DELETION TRACKING [1 indicator]

Deleted file recovery opportunities

- Partition 0: Orphaned files found (deleted but recoverable): 1 entries

MFT ANOMALIES [1 indicator]

Master File Table irregularities

- Partition 0: MFT contains 4 system file entries

SUMMARY STATISTICS

Category	Indicators
Timestamping	256
Shadow Copy Deletion	0
Hidden Streams (ADS)	0
Log Clearing	2
Registry Tampering	0
File Deletion	1
MFT Anomalies	1
TOTAL	260

Overall Anti-Forensic Risk Level: HIGH

RECOMMENDATIONS

1. Verify the integrity of the USN journal and system backup mechanisms
2. Investigate potential timestamping and shadow copy deletion tools used
3. Verify the file's history and investigate potential deletion tools used

Additional Anti-Forensic Investigation Steps:

- Conduct detailed timeline analysis around detected anomalies.
- Examine Windows Event Logs for evidence of vssadmin or wevtutil commands.

- Analyze \$MFT and \$UsnJrnl for file system manipulation evidence.
- Review Registry hives for persistence mechanisms and tool artifacts.
- Search for known anti-forensic tool signatures and artifacts.